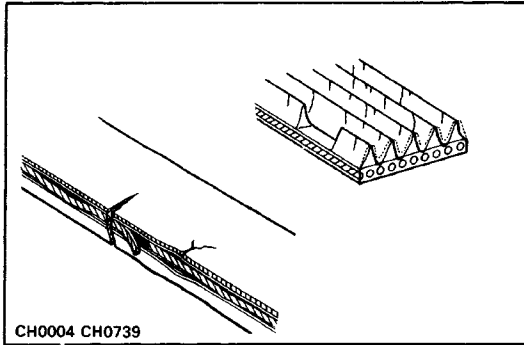




3. INSPECT DRIVE BELT

- (a) Visually check the drive belt for excessive wear, frayed cords etc.

If necessary, replace the drive belt.

HINT: Cracks on rib side of a drive belt are considered acceptable. If the drive belt has chunks missing from the ribs, it should be replaced.

- (b) Using a belt tension gauge, measure the drive belt tension.

Belt tension gauge:

Nippondenso BTG-20 (95506-00020)

Borroughs No. BT-33-73F

Drive belt tension:

4A-FE		New belt 160 ± 20 lbf
		Used belt 130 ± 20 lbf
3S-GTE	w/ A/C	New belt 165 ± 10 lbf
		Used belt 115 ± 20 lbf
	w/o A/C	New belt 150 ± 25 lbf
		Used belt 130 ± 25 lbf
	w/ A/C	New belt 165 ± 10 lbf
		Used belt 110 ± 10 lbf
5S-FE	w/o A/C	New belt 125 ± 25 lbf
		Used belt 95 ± 20 lbf

If the belt tension is not as specified, adjust it.

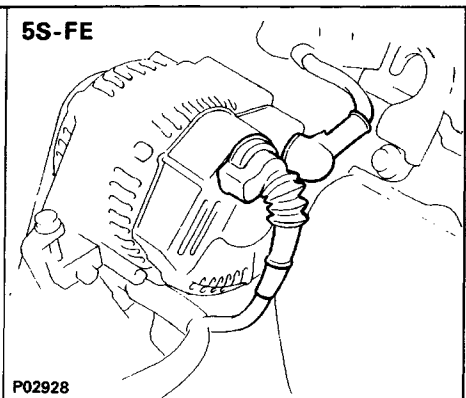
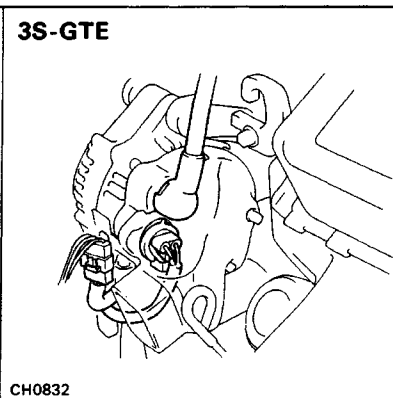
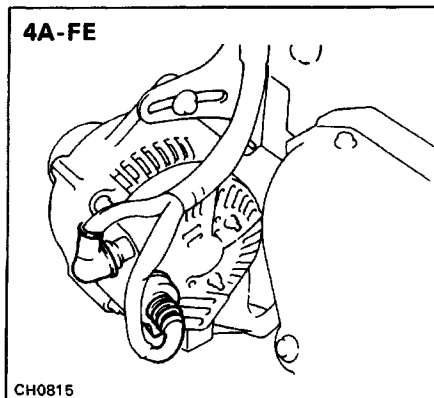
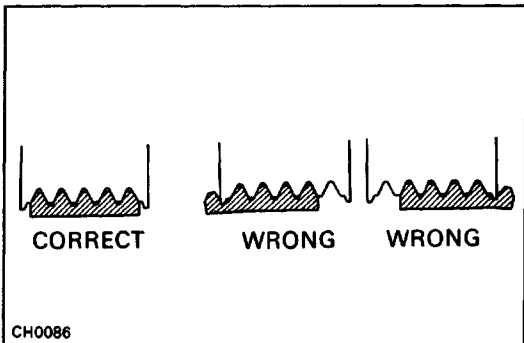
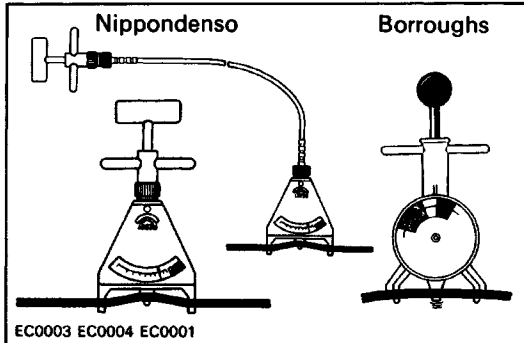
HINT:

- "New belt" refers to a belt which has been used less than 5 minutes on a running engine.
- "Used belt" refers to a belt which has been used on a running engine for 5 minutes or more.
- After installing a belt, check that it fits properly in the ribbed grooves.
- Check by hand to confirm that the belt has not slipped out of the groove on the bottom of the pulley.
- After installing a new belt, run the engine for about 5 minutes and recheck the belt tension.

4. VISUALLY CHECK GENERATOR WIRING AND LISTEN FOR ABNORMAL NOISES

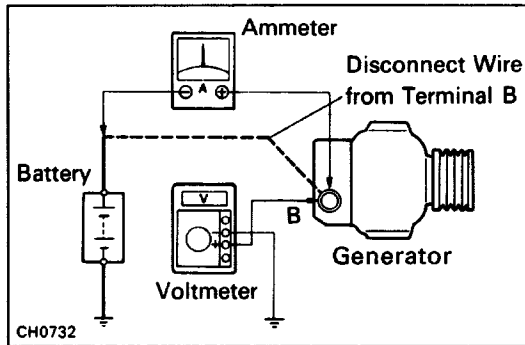
- (a) Check that the wiring is in good condition.

- (b) Check that there is no abnormal noise from the generator while the engine is running.

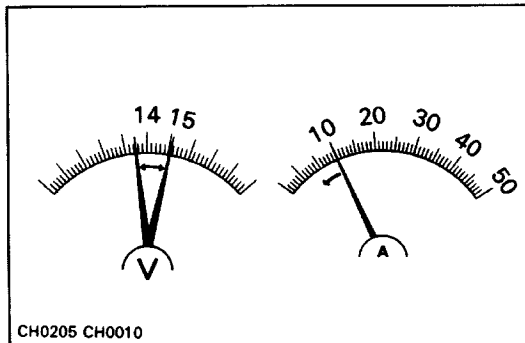


5. INSPECT CHARGE WARNING LIGHT CIRCUIT

- Turn the ignition switch ON. Check that the charge warning light is lit.
- Start the engine. Check that the light goes off.
If the light does not go off as specified, troubleshoot the charge light circuit.



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6. INSPECT CHARGING CIRCUIT WITHOUT LOAD

HINT: If a battery/generator tester is available, connect the tester to the charging circuit as per manufacturer's instructions.

- If a tester is not available, connect a voltmeter and ammeter to the charging circuit as follows:
 - Disconnect the wire from terminal B of the generator and connect it to the negative probe of the ammeter.
 - Connect the positive (+) probe of the ammeter to terminal B of the generator.
 - Connect the positive (+) probe of the voltmeter to terminal B of the generator.
 - Ground the negative (–) probe of the voltmeter.
- Check the charging circuit as follows:
With the engine running from idle to 2,000 rpm, check the reading on the ammeter and voltmeter.

Standard amperage: 10 A or less

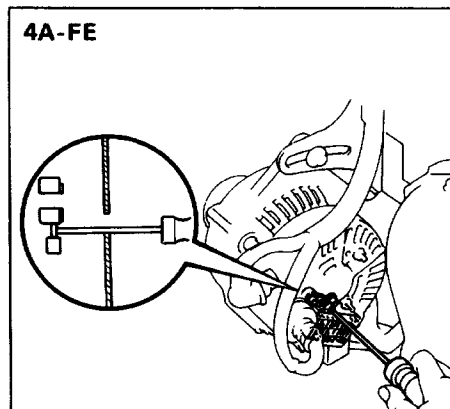
Standard voltage: 13.9 – 15.1 V at 25°C (77°F)

13.5 – 14.3 V at 115°C (239°F)

If the voltmeter reading is greater than standard voltage, replace the IC regulator.

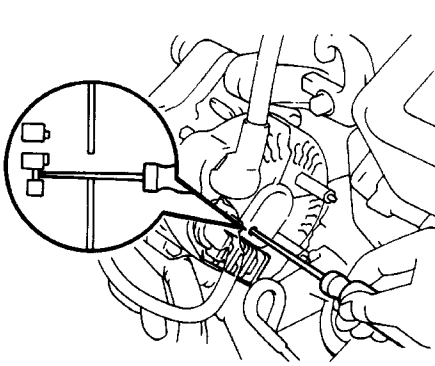
If the voltmeter reading is less than standard voltage, check the IC regulator and generator as follows:

- With terminal F grounded, start the engine and check the voltmeter reading of terminal B.



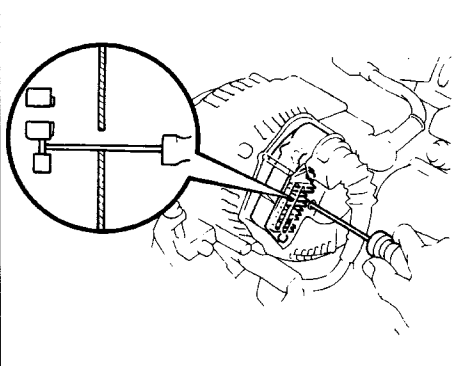
CH0814

3S-GTE

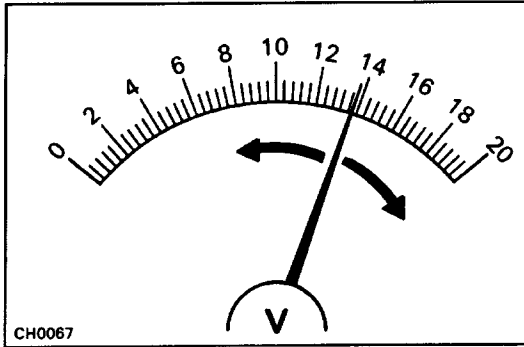


CH0833

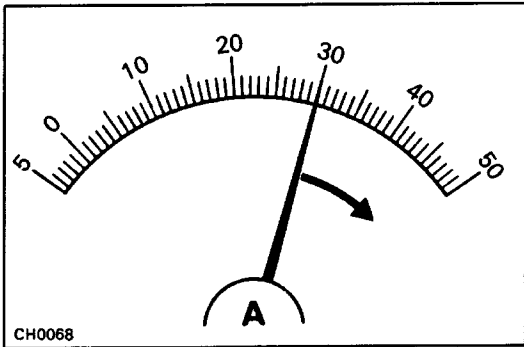
5S-FE



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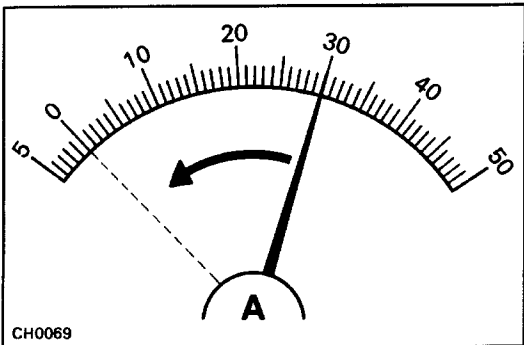
- If the voltmeter reading is greater than standard voltage, replace the IC regulator.
- If the voltmeter reading is less than standard voltage, check the generator.



7. INSPECT CHARGING CIRCUIT WITH LOAD

- With the engine running at 2,000 rpm, turn on the high beam headlights and place the heater blower switch at "HI".
- Check the reading on the ammeter.

Standard amperage: 30 A or more



If the ammeter reading is less than the standard amperage, repair the generator. (See page [CH-12](#))
HINT: With the battery fully charged, the indication will sometimes be less than standard amperage.